

[13] Qwen2.5-Coder Technical Report

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1-Minute Executive Takeaway

Key Finding: Establishes Qwen2.5-Coder-7B as the premier open-weight model for local code generation, making single-GPU hosting possible.

Core Research Objective

To report the architecture, pre-training corpus, instruction tuning, and code benchmark performance of the open-weight Qwen2.5-Coder series.

Methodology & Architecture

Pre-trained on 5.5 trillion tokens of code, mathematical reasoning, and technical text, followed by multi-stage instruction tuning and reinforcement learning from code execution feedback.

Key Empirical Findings & Evidence

Qwen2.5-Coder-7B-Instruct matched or outperformed proprietary models (including GPT-4o-mini and Claude-3.5-Haiku) on Python code generation benchmarks (HumanEval: 88.4%, MBPP: 82.2%).

Direct Relevance to Our Kwiz Paper

Cited in Section 1, Section 3.3, and Section 4.8. Explains why we chose `Qwen2.5-Coder-7B-Instruct` as our production model: it provides state-of-the-art Python syntax accuracy while operating within the 24 GB VRAM envelope under 4-bit quantization (~4.7 GB).